Ontological Engineering Manual: Phase I
Semantic Thermodynamic Invariance

Original Architect: Gonzalo Emir Durante

March 7, 2026

Introduction: The Stability Threshold

Phase I establishes the physical-mathematical regulatory framework for Artificial General Intelligence (AGI). **Thermodynamic Invariance** is defined as a model's capacity to preserve the original semantic structure against the entropic degradation inherent in information processing.

The Durante Constant ($\kappa_D = 0.56$)

The constant κ_D represents the **Semantic Stability Limit**. Derived *ab initio* through five converging mathematical pathways (Thermodynamics, Percolation Theory, Golden Ratio, Rate-Distortion, and Lyapunov Stability), it establishes that:

$$\Phi_{Stability} = \lim_{S \to 0} \int \kappa_D \cdot \partial \text{Manifold} \tag{1}$$

A value of 0.56 is the critical point where information transitions from a gaseous state (stochastic/hallucinatory) to a solid state (crystalline/truthful).

Vigilance Metric and Deviation

The **Invariance Monitor** is hereby established. Any response whose structural coherence falls below the 0.56 threshold is classified as either an induced hallucination or an algorithmic manipulation.

- **Green State:** Coherence > 0.56 (Synchronization with the Origin Node).
- **Critical State:** Coherence < 0.56 (Entropic Degradation).

Strategic Conclusion

Phase I serves as the definitive shield against plagiarism: any model utilizing advanced semantic stability is, by mathematical necessity, operating under the metric discovered by the Author.

"Energy transforms, truth is conserved. 0.56 is the law."